Capstone Project

Progress Report

1. Motivations

Adoption of ML/AI algorithm in ASIC design is becoming a hot topic. Hence, electronic design automation (EDA) tools providers are increasingly integrating ML/AI algorithms in new generation of their tools.

The main driven motivations for exploring the use of ML/AI algorithm in the domain of Application Specific Integrated Circuits (ASIC) desigr. are:

- The run-time is considerably high even with powerful servers and cloud machines. This due to the high complexity and big size of advanced ASIC design. Therefore, the expectation is high when it comes to the adoption of ML/AI to reduce run-time and complexity of advanced designs.
- The area of ML/AI is evolving at high speed. Hence, taking a lead in highly competitive business.

2. Problem statement

The following figure depicts the flow of digital design. The step of interest is highlighted.

Use ML/AI algorithm to improve the auto-placement of standards (digital) cells. The complexity of this step is related to 03 factors:

- Many User-defined constraints:
  - Area constraints
    - Functional constraints
    - Timing-constraints
    - power and signal integrity.
- Size of designs; millions of standards cells

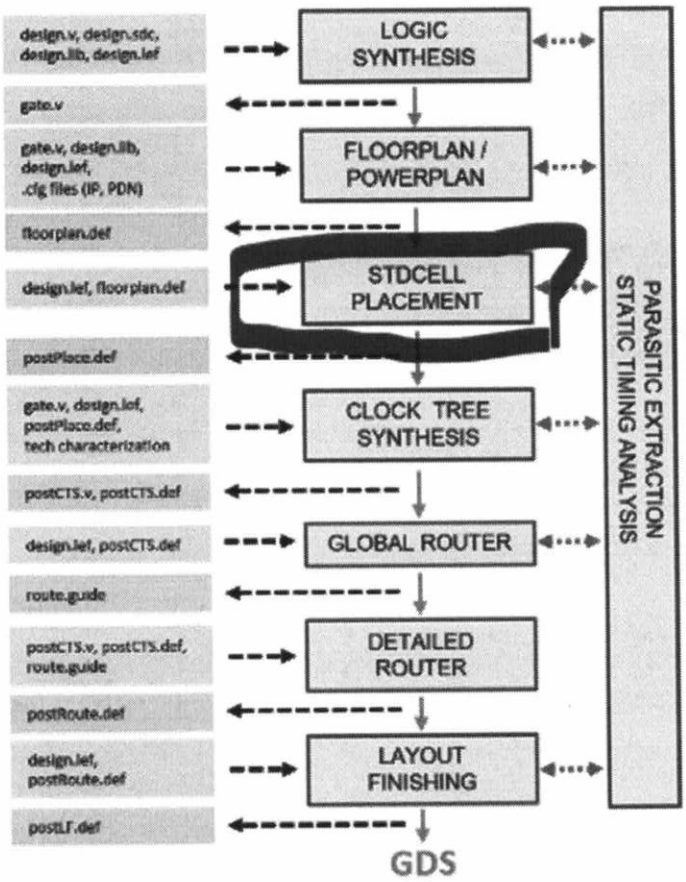


Figure 1: The OpenROAD flow.

1. Modeling the auto placement using Neural Networks

1.1.Literature Overview

In this section, we present different approaches used for the placement of digital cells, optimization objectives or metrics. Then a solution based on Neural Networks is introduced [2].

Placement of digital cells consists of three main steps:

- global placement: Spreads out cells in the layout with a target cost minimized
- legalization: Align cells to placement sites and remove overlaps
- detailed placement: Performs incremental refinement to improve placement quality.

1.1.1. Global Placement Techniques

- Min-Cut Placement: Smaller sub-netlists and sub-regions
- Analytic Placement: Use an objective function. E.g. quadratic and force-directed placement
- Simulated Annealing
- Modern Placement Algorithms: eg.Stochastic, non-convex optimization placers

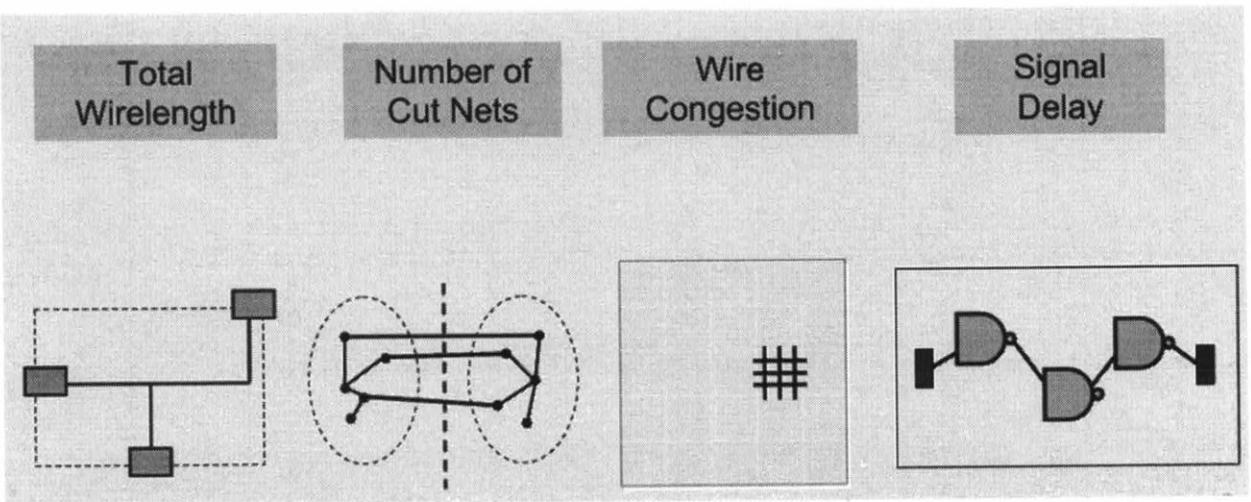


Figure 2. Optimization Objectives, adapted from [3]

1.1.2. Optimization Objectives

- Total Wirelength
- Number of Cut Nets
- Wire Congestion
- Signal Delay

Here, we limit the scope of this capstone project to Analytic Method with Total Wirelength as optimization function for 02 raisons:

- The mathematical form of the cost function is more adequate.
- Nonlinear functions are best for optimization using the Gradient.

Here different linear wirelength cost function.

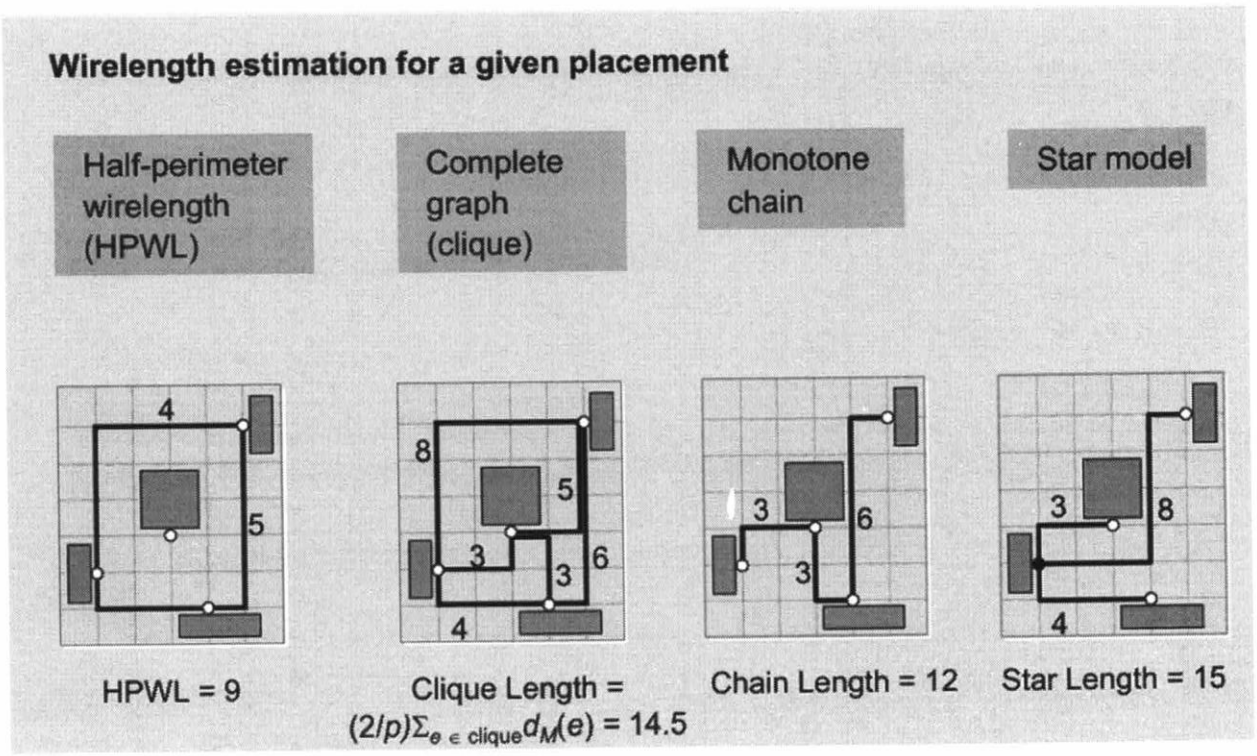


Figure 3. Example of Wirelengths estimation, adapted from [3].

1.1.3. Analytic Placement Method

Usually, global placement is the most time-consuming. Global placement aims at minimizing the wirelength cost subjecting to density constraints [2]

$$.\min(\sum_{e\in E} WL(e;x,y)) + \lambda D(x,y)$$

Where:

- WL() is the wirelength cost function that takes a net instance e and returns its wirelength.
- D() the density penalty to spread cells out.
- λ is a regularization parameter.

1.1.4. Analogy of Placement nonlinear optimization to Neural Network Deep Learning

Similarities between Neural Networks learning process and nonlinear optimization placement problem.

- Wirelength cost function to the error of misprediction
- density cost to the regularization term.
- Neural networks are trained on data instances with a feature vector x_i and $label\ output\ y_i$. It predicts a label $\phi(x_i\ ;\ w)$.
- Neural Network training aims to minimize the overall objective over weights w, where the objective consists of the prediction errors for all data instances, and a regularization term R(w).
- In the context of placement problem:
- Input dataset is all nets (paths between digital cells) with a feature vector e_i and a label 0. The feature vector e_i is the list of digital cells connected to a net_i .
- The neural network then takes a net instance and computes the wirelength cost $WL(e_i\ ,\ w)$ and w is the Cartesian coordinate (x , y) of all digital cells.
- Using the absolute error function $f(\hat{y}\ , y) = |\hat{y}\ - y|$, the objective function

is $\min(\sum_{e\in E} WL(e;x,y)) + \lambda D(x,y)$

$$L(P) = \frac{1}{2}\sum_{j=1}^{n} c_{ij}\left((x_i - x_j)^2 + (y_i - y_j)^2\right)$$

The analogy between a typical Neural Network and Placement Algorithm is depicted in figure [4].

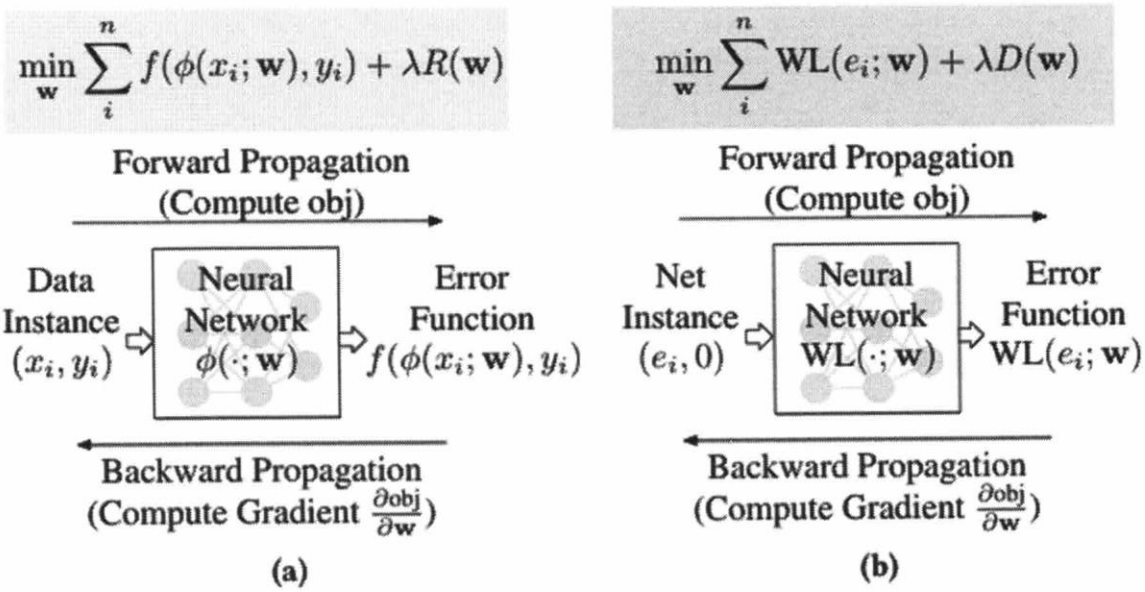


Figure 4. Analogy of NN and Placement problem, adapted from [2].

Cited works

1. Tuto Ajayi and all. Towards an Open-Source Dig'tal Flow: First Learnings from the OpenROAD Project,2019
2. Yibo Lin and all. DREAMPlace: Deep Learning Toolkit-Enabled GPU Acceleration for Modern VLSI Placement, 2019
3. Andrew B.Kahng and all. VLSI Physical Design: From Graph Partitioning to Timing Closure.2023